

Certified Polyhedral Projection for Counterfactual Explanations

Generating Valid, Proximal, and Robust Counterfactuals
via Preimage Approximation

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Outline

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What Are Counterfactual Explanations?

A classifier labels an input x_0 as class l . We ask:

Goal

Find x' classified as a **different target class** t , as close as possible to x_0 .

Example (MNIST): “What minimal change turns a **digit 1** into a **digit 7**?”

Desirable properties:

- ① **Validity** – guaranteed to be classified as the target class
- ② **Proximity** – minimal distance to the original input
- ③ **Plausibility** – stays close to the data manifold
- ④ **Robustness** – remains valid under small perturbations

Why Not Gradient Descent?

Standard methods minimize a loss that trades off distance and target-class confidence:

Problems

- **Non-convex** – gets stuck in local minima
- **No validity guarantee** – result may sit right at the decision boundary
- **Not robust** – tiny perturbations can flip the prediction back

Our approach: replace gradient descent
with **projection onto certified safe regions**.

Key Insight

Around each training sample we can build a **convex region** (polytope) where the classifier is *guaranteed* to predict a specific class.

Finding a counterfactual = **projecting onto the nearest such region**.

- Projection onto a convex polytope is a **convex optimization problem**
- Convex $\Rightarrow$ **global optimum**, no local minima
- No trade-off hyperparameter λ needed

Two-Phase Architecture

Phase I – Offline: Build Atlas

For each training sample $\mathbf{c}_i$ of class t :

- 1 Consider a small region around $\mathbf{c}_i$ (an ε -ball)
- 2 Use **LiRPA** to find linear constraints that *certify* the classifier predicts t throughout this region
- 3 Store the resulting **certified polytope** $\mathcal{P}_i$

$\Rightarrow$ Every point inside $\mathcal{P}_i$ is **guaranteed class** t

Phase II – Online: Query

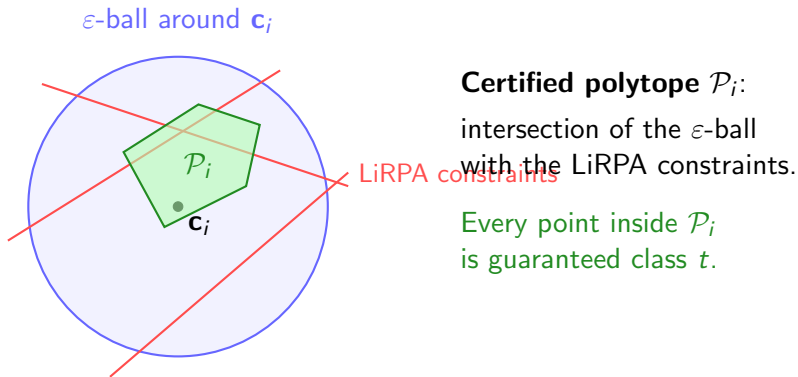
Given a query $\mathbf{x}_0$:

- 1 Use a **spatial index** (BVH) to quickly find nearby polytopes
- 2 **Project** $\mathbf{x}_0$ onto each candidate polytope (convex QP)
- 3 Return the **closest** valid projection

$\Rightarrow$ Result is **certifiably** classified as target

Step 1: Certifying Regions with LiRPA

LiRPA (Linear Relaxation based Perturbation Analysis) gives us **linear lower bounds** on the network within a small region around each sample.



The collection of all polytopes for a class forms the **Certified Atlas**.

Step 2: Fast Search with BVH

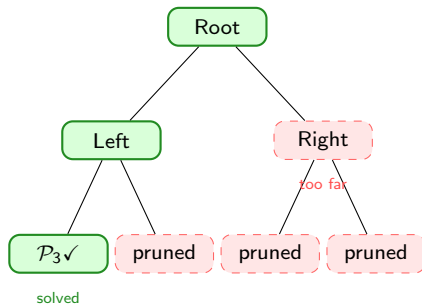
With ~ 1000 polytopes per class, checking all of them is too slow.

Bounding Volume Hierarchy

- Organize polytopes in a **binary tree** of bounding boxes
- **Branch-and-bound**: skip subtrees that can't contain a closer match
- Same result as exhaustive search

Result on MNIST ($d = 784$):

- Exhaustive: ~ 1000 projections/query
- BVH: $\sim 2\text{--}5$ projections/query
- $200\text{--}500\times$ **speedup**

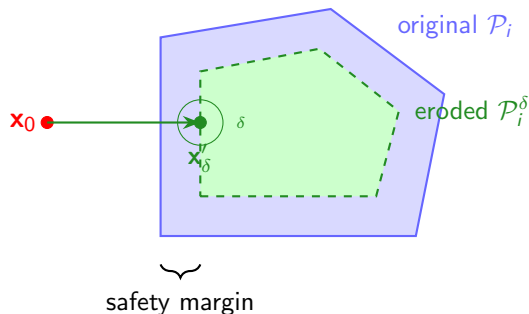


Making Counterfactuals Robust

A counterfactual at the polytope boundary can be flipped by a tiny perturbation.

Idea: Erode the Polytope Inward by δ

Shrink constraints so that the **entire ball of radius δ** around the counterfactual stays inside.



- Larger $\delta \Rightarrow$ more robust, but farther from original
- Too large $\delta \Rightarrow$ eroded polytope becomes empty

Cross-Norm Robustness

The norm used to build the atlas (p) and the robustness norm (q) can **differ**.

Two things must be adjusted:

1. **Each linear constraint** is pushed inward by $\delta \cdot \|\text{normal}\|_{q^*}$, where q^* is the **dual** of q :

Robustness norm q	L_1	L_2	L_∞
Dual norm q^*	L_∞	L_2	L_1

2. **The ε -ball** must shrink to accommodate the robustness ball. The cost depends on a **key norm inequality**:

$$\|\mathbf{x}\|_{p_2} \leq \|\mathbf{x}\|_{p_1} \quad \text{for } p_2 > p_1$$

- If $q < p$ (e.g., L_2 robustness on L_∞ atlas): the robustness ball **fits for free** $\Rightarrow$ **cheap**
- If $q > p$: the ball must be shrunk by a factor growing with dimension $d \Rightarrow$ **expensive in high d**

Results on MNIST

Setup:

- CNN classifier, 99.2% accuracy
- 10,000 polytopes (1000/class)
- 784-dimensional input

Counterfactual generation:

- **100% validity** (by construction)
- BVH: **2–5 projections/query**
(vs. 1000 exhaustive)
- 180 counterfactuals evaluated

Metric	Mean
ℓ_2 distance	23.4
Pixels changed (/784)	224
Sparsity	29%
LOF score	1.55
QP solves (BVH)	3.4

- $\text{LOF} \approx 1$: counterfactuals look like **real target-class data**
- $\sim 29\%$ pixels changed: **moderately sparse**

Summary

What we did

- ① **Certified Polyhedral Projection** – counterfactual generation via convex projection instead of gradient descent
- ② **Provable validity** – every counterfactual lies inside a certified region
- ③ **Robustness** – polytope erosion guarantees stability under perturbations, even across different norms
- ④ **Efficiency** – BVH spatial indexing gives $200\text{--}500\times$ speedup

Key advantages over standard methods

- No trade-off hyperparameter λ
- Convex QP $\Rightarrow$ global optimum per polytope
- Atlas built once offline, queries are fast

Thank You

Questions?

`github.com/gabrielepintus/PreimageCounterfactualSampling`